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AUTNES Automatic Content Analysis of the Media Coverage 2019

Documentation

Version 1.1

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Introduction

This Documentation Report accompanies and describes the data of the AUTNES Automatic Content Analysis of Media Coverage of the Austrian National Election held on 29 September 2019. The body elected was the National Parliament (Nationalrat).

The Documentation Report is structured in six sections. The first section outlines the conditions of use of the data. The second section, the study description, provides an overview of the study. The third section presents the study design, i.e., the media sample and the sampling procedure. The fourth section provides information about the coding procedure, presenting all the variables and the units of analysis. The sixth section describes the validation procedure. The last section provides an overview of the frequencies of all variables in the analyzed media outlets. Precise information about the coding instructions for the validation procedure can be found in the Validation Codebook (in German), which is also available via www.aussda.at.

Acknowledgement of the data

Data users are required to acknowledge the data and the accompanying release document. Please refer to the AUSSDA data catalogue (https://data.aussda.at, doi: 10.11587/ZY7KSQ) for a recommendation on how to cite these data and the documentation.

1. Conditions of Use

1.1. Restrictions

The data are available under the terms listed at the AUSSDA Dataverse.

1.2. Confidentiality

AUTNES, the Principal Investigators and the funding institution bear no responsibility for the use of the data, or for interpretations or inferences based on their use, neither do they accept liability for indirect, consequential or incidental damages or losses arising from the use of the data.

1.3. Deposit Requirement

To facilitate exchanges within the scientific community and to provide funding agencies with essential information about the use of archival resources, users of the AUTNES data are requested to notify the AUTNES team of all forms of publications referring to AUTNES data.

2. Study description

2.1. Title and Edition

AUTNES Automatic Content Analysis of the Media Coverage 2019

2.2. Principal Investigators

Wolfgang C. Müller, University of Vienna (Supply Side)

Sylvia Kritzinger, University of Vienna (Demand Side)

Hajo G. Boomgaarden, University of Vienna (Media Side)

2.3. Funding and Acknowledgements

The study was carried out under the auspices of the ACIER - Austrian Cooperative Infrastructure for Electoral Research and VieCER - Vienna Center for Electoral Research, financed by the Austrian Federal Ministry of Education, Science and Research. This data collection has been supported by the Austrian Press Agency (APA) via the APA-UNIVIE Data Project.

2.4. Data File Names

10754_da_en_v1_0.dta	Core data file - STATA format - 57 Variables, 44310 Observations
10754_da_en_v1_0.csv	Core data file - CSV format – 57 Variables, 44310 Observations
10754_da_en_v1_0.sav	Core data file - SPSS format – 57 Variables, 44310 Observations

2.5. Topics /Keywords

Election coverage, Political actors, Mass media coverage, Elections, Election campaigns, Political parties, Political representation, Political issues, European Union, Content Analysis

3. Study design

The data set is based on the automatic analysis of all political news reports published in Austrian media outlets (print, online, TV, radio) leading up to the Austrian National Election 2019. We define a news report as political when the report mentions a central political actor. Central political actors are

- designations such as government (coalition, governing parties) and opposition (opposition parties) on the national level
- members of the government before the election, on national, state and EU level
- ministries and state secretaries (Staatssekretäre) on the national level
- the Austrian President
- members of parliament before the election, both national and EU level, or the parliament as an institution
- Austrian political parties eligible on the national level, including demographically defined sub-organizations on the national level (e.g. for women)
- directorate members of demographically defined sub-organizations of Austrian parties (also political fractions in unions)
- political candidates running for parliament or office in the election

If any of these actors is mentioned in a news item, it was considered political news. The period of analysis ranges from 31 May to 29 September 2019 for TV, radio, and online media, and from 31 August to 29 September 2019 for print media. Note that the time and media range of news reports extends the range covered by the AUTNES manual content analysis, which only includes print media. The media data can be synced with AUTNES demand side data, specifically with survey waves 10 and 11 of the AUTNES Online Panel

Study 2017-2019 (see AUTNES Online Panel Study 2017-2019, doi: 10.11587/QDETRI for more information on and documentation for the survey data).

3.1. Media Sample

The media sample was selected based on the type and circulation to reflect the Austrian national media landscape. All media reports that mentioned a relevant political actor were downloaded from the APA (Austrian Press Agency) database. Political actors in that sense are all members of the government, all lead candidates, and the parties themselves that were represented in the parliament before or after the elections. The data includes reports from 26 TV, radio, print and online media outlets, totaling 44,310 individual reports. For an overview see *Table 1*. The table also includes information on whether a media outlet was already included in the AUTNES Automatic Content Analysis of Media Coverage from 2013 and 2017. Some media outlets were only partially included. For example, whereas this data set includes reports from *ZIB 2 am Sonntag*, this specific news broadcast was not included in the *ZIB 2* data in 2013 and 2017.

Table 1. Media sample and characteristics of media outlets.

Media outlet	Time	Media	Type	AUTNES17	AUTNES13	N
ZIB 1	31.05.-29.09.	Public Service TV	TV	Yes	Yes	559
ZIB 2	31.05.-29.09.	Public Service TV	TV	Partial	Partial	404
ZIB (other)	31.05.-29.09.	Public Service TV	TV	Partial	No	1346
Regional News (TV)	31.05.-29.09.	Public Service TV	TV	No	No	75
ATV	31.05.-29.09.	Commercial TV	TV	Partial	Yes	100
Puls 4	31.05.-29.09.	Commercial TV	TV	Yes	No	86
Der Standard	31.08.-29.09.	Quality Print	Print	Yes	Yes	846
Die Presse	31.08.-29.09.	Quality Print	Print	Yes	Yes	689
Salzburger Nachrichten	31.08.-29.09.	Quality Print	Print	Yes	Yes	580
Kronen Zeitung	31.08.-29.09.	Tabloid Print	Print	Yes	Yes	1868
Österreich	31.08.-29.09.	Tabloid Print	Print	Yes	Yes	1278
Heute	31.08.-29.09.	Tabloid Print	Print	Yes	Yes	398
Kurier	31.08.-29.09.	Mid-market Print	Print	Yes	Yes	1371
Kleine Zeitung	31.08.-29.09.	Mid-market Print	Print	Yes	Yes	1308
derstandard.at	31.05.-29.09.	Quality online	Online	Yes	Yes	5787
diepresse.com	31.05.-29.09.	Quality online	Online	Yes	Yes	1204
sn.at	31.05.-29.09.	Quality online	Online	Yes	Yes	1595
krone.at	31.05.-29.09.	Tabloid online	Online	Yes	Yes	5613

oe24.at	31.05.-29.09.	Tabloid online	Online	Yes	Yes	633
heute.at	31.05.-29.09.	Tabloid online	Online	Yes	Yes	2453
orf.at	31.05.-29.09.	Mid-market online	Online	Yes	Yes	2880
kurier.at	31.05.-29.09.	Mid-market online	Online	Yes	Yes	8964
kleinezeitung.at	31.05.-29.09.	Mid-market online	Online	Yes	Yes	1667
Ö1	31.05.-29.09.	Public Service Radio	Radio	No	Partial	1933
Regional News (Radio)	31.05.-29.09.	Public Service Radio	Radio	No	No	245
Ö3	31.05.-29.09.	Public Service Radio	Radio	No	No	428

3.2. Sample of News Items

To collect the data, articles for print newspapers were downloaded from the database of the Austria Presse Agentur (APA, www.apa-defacto.at), using the following search string containing the names of all Austrian political parties participating in the National Election of 2019, the names of the chancellor, all ministers and secretaries of state (from the cabinets Kurz I, Löger as well as Bierlein), as well as the national parliament as an institution. The search string that was used to retrieve those articles was:

```
nationalrat* OR övp* OR "liste kurz"~4 OR "neue* volkspartei" OR spö* OR fpö* OR neos* OR "liste pilz"~4 OR "liste jetzt*"~4 OR "rot-grün" OR "schwarz-grün" OR (("die grüne*" OR "der grüne*" OR "den grünen") NOT ("grün* rasen"~2 OR "grün* wiese*"~2)) OR ökopartei OR grünpolitiker* OR "grün* politiker*"~4 OR "tiroler grüne*" OR "grüne* tirol" OR "wiener grüne*" OR "grüne* wien" OR "grüne* burgenland" OR "burgenländ* grüne*" OR "grüne* kärnten" OR "kärntner* grüne*" OR "grüne* salzburg" OR "salzburger* grüne*" OR "grüne* oberösterreich" OR "oberösterreichische* grüne*" OR "grüne* steiermark" OR "steirische* grüne*" OR "grüne* vorarlberg" OR "vorarlberger grüne*" OR "grüne* niederösterreich*" OR "niederösterreichische* grüne*" OR "kanzler* kurz"~5 OR "bundeskanzler* kurz"~5 OR "kurz volkspartei"~5 OR "sebastian kurz"~5 OR "kurz vp"~5 OR "rendi-wagner" OR "pamela rendi"~5 OR "pamela wagner"~5 OR "norbert hofer"~5 OR "hofer fp"~5 OR "hofer freiheitlich*"~5 OR "meinl-reisinger" OR "beate meinl"~5 OR "beate reisinger"~5 OR "peter pilz"~5 OR "politiker* pilz" OR "abgeordnet* pilz"~5 OR "parlament* pilz"~5 OR "aufdecker* pilz"~5 OR "maria stern"~5 OR "werner kogler"~5 OR "spitzenkandi* kogler"~5 OR "kogler grün*"~5 OR "kogler eu"~5 OR bierlein OR jabloner OR "eduard müller"~5 OR "minister* müller"~5 OR "finanzminister* müller"~5 OR "bundesminister* müller"~5 OR schallenberg OR zarfl OR rauskala OR "udolf-strobl" OR "minister* udolf"~5 "bundesminister* udolf"~5
```

"wirtschaftsminister* udolf"~5 OR peschorn OR starlinger OR "maria patek"~5 OR "minister* patek"~5 OR "bundesminister* patek"~5 OR "nachhaltigkeit* patek"~5 OR "tourismus* patek"~5 OR "landwirtschaft* patek"~5 OR "andreas reichhardt"~5 OR "minister* reichhardt"~5 OR "bundesminister* reichhardt"~5 OR "infrastrukturminister* reichhardt"~5 OR "verkehrsminister* reichhardt"~5 OR "ines stilling"~5 OR "minister* stilling"~5 OR "bundesminister* stilling"~5 OR "frauenminister* stilling"~5 OR "familienminister* stilling"~5 "kanzler löger"~5 OR "bundeskanzler löger"~5 OR "eckart ratz"~5 OR "minister* ratz"~5 OR "bundesminister* ratz"~5 OR "innenminister* ratz"~5 OR "walter pöltner"~5 OR "minister* pöltner"~5 OR "bundesminister* pöltner"~5 OR "sozialminister* pöltner" OR "arbeitsminister* pöltner"~5 OR "gesundheitsminister* pöltner"~5 OR "johann luif"~5 OR "minister* luif"~5 OR "bundesminister* luif"~5 OR "verteidigungsminister* luif"~5 OR "heeresminister* luif"~5 OR "valerie hackl"~5 OR "minister* hackl"~5 OR "bundesminister* hackl"~5 OR "verkehrsminister* hackl"~5 OR "infrastrukturminister* hackl"~5 OR strache OR "hartwig löger"~5 OR "vizekanzler* löger"~5 OR "minister* löger"~5 OR "bundesminister* löger"~5 OR "finanzminister* löger"~5 OR "bogner-strauß" OR "bundesminister minister* bogner*"~5 OR "bundesminister* bogner*"~5 OR "minister* strauß*"~5 OR "bundesminister* strauß*"~5 OR "karin kneissl"~5 OR "minister* kneissl"~5 OR "bundesminister* kneissl"~5 OR "außenminister* kneissl"~5 OR kickl OR "josef moser"~5 OR "minister* moser"~5 OR "bundesminister* moser"~5 OR "justizminister* moser"~5 OR "reformminister* moser"~5 OR "hartinger-klein" OR "minister* hartinger*"~5 OR "bundesminister* hartinger*"~5 OR "sozialminister* hartinger*" OR "arbeitsminister* hartinger*"~5 OR "gesundheitsminister* hartinger*"~5 OR köstinger OR kunasek OR "norbert hofer"~5 OR "minister* hofer"~5 OR "bundesminister* hofer"~5 OR "infrastrukturminister* hofer"~5 OR "verkehrsminister* hofer"~5 OR "heinz faßmann"~5 OR "minister* faßmann"~5 OR "bundesminister* faßmann"~5 OR "bildungsminister* faßmann"~5 OR "margarethe schramböck"~5 OR "minister* schramböck"~5 OR "bundesminister* schramböck"~5 OR "wirtschaftsminister* schramböck" OR "gernot blümel"~5 OR "minister* blümel"~5 OR "blümel vp"~5 OR "blümel volkspartei*"~5 OR "bundesminister* blümel"~5 OR "kanzleramt* blümel"~5 OR "kulturminister* blümel"~5 OR edtstadler OR "hubert fuchs"~5 OR "staatssekretär* fuchs"~5 OR "van der bellen" OR bundespräsident* OR expertenregierung* OR expertenminister* OR beamtenregierung*

Articles were deduplicated based on text, title, date, medium (all of those had to be the same to be considered a duplicate). The data collection and the subsequent cleaning resulted in a final data set consisting of 44,310 articles. For further analysis, the data set was uploaded to the online content analysis platform AmCAT – see www.amcat.nl for more information and documentation.

4. Coding Procedure

4.1. Overview

We used Lucene syntax search strings to measure all variables. Wherever possible, we built on already developed search strings from the AUTNES 2013 and 2017 Automatic Content Analysis. These were available for all parties included in our data set – although the search string for JETZT had to be adapted after the renaming of Pilz List – two main candidates (Sebastian Kurz, Peter Pilz), most style variables and all issues (for a complete overview of available strings see the documentation of AUTNES 2013 and 2017 Automatic Content Analysis). New search strings were created for all new variables as well as to extend our issue search strings with issues arising after the 2017 election (e.g., climate strikes). These search strings were then adapted, iterated on, and constantly tested against a set of articles. For sentiment analysis, we made use of two already existing dictionaries on sentiments in German language by Rauh and SentiWS.

For all variables except the issue and sentiment variables, the adaption procedure included adding and removing of search terms, adapting the word distances, and varying different sub-blocks and operators. After each adaption round, the search strings were tested for recall and precision against a set of 100 articles. For the most complex search strings conflict and games framing) the R-package *quanteda* (Benoit et al., 2018[1]) was used to retrieve all words in a window of 5 words before and after the original search-terms. Those words were then listed in order of occurrence and the most frequent ones were added to

[1] Benoit, K., Watanabe, K., Wang, H., Nulty, P., Obeng, A., Müller, S., & Matsuo, A. (2018). quanteda: An R package for the quantitative analysis of textual data. *Journal of Open Source Software*, *3*(30), 774. https://doi.org/10.21105/joss.00774

the search string. This procedure was repeated until no significant change in performance (i.e., recall and precision) was achieved.

For the issue variables, more than 800 search strings were tested against an initial validation set. The results were then aggregated to the highest level (see Table 2 for an overview of the highest issue levels). A sample of false positives as well as false negatives on the aggregated level were checked both manually and automatically for similarities (sample n=100 for each issue). Following this, new search strings were created, and already existing search strings adapted.

Finally, we checked the sentiment dictionaries for possible confounders. This meant going manually through every word or word combination included in the dictionaries and controlling for possible overlaps with prominent candidate and party names in the 2019 national election. This procedure led to the removal of the word "*kurz*" from the dictionary by Rauh due to possible conflicts with the appearance of Sebastian Kurz, the main candidate of the ÖVP, in media reports.

The search strings as well as their recall and precision values are presented in the next subsections. A detailed account of the final validation procedure can be found in the following section (Section 5). These strings together with the meta data of the articles were used to code all variables on article level.

*Table **2*** provides an overview of all 55 variables included in the data set.

Table 2. Variables overview.

Variable	Label	Type
v0	Article ID	Metadata
v1	Headline	Metadata
v2_l1	Medium Level 1	Metadata
v2_l2	Medium Level 2	Metadata
v2_l3	Medium Level 3	Metadata
v2_l4	Medium Level 4	Metadata
v3_13	Medium Included: AUTNES13	Metadata
v3_17	Medium Included: AUTNES17	Metadata
v4	Date	Metadata
v5	Page	Metadata
v6	Section	Metadata
v7	Word count	Metadata
v8a	SPÖ	Party
v8b	ÖVP	Party
v8c	FPÖ	Party
v8d	Greens	Party
v8e	NEOS	Party
v8f	JETZT - Pilz List	Party
v9a	Pamela Rendi-Wagner	Candidate
v9b	Sebastian Kurz	Candidate
v9c	Norbert Hofer	Candidate
v9d	Werner Kogler	Candidate
v9e	Beate Meinl-Reisinger	Candidate
v9f	Peter Pilz	Candidate
v10	Government of Experts	Government
v11a	Economy, Budget and Taxes	Issue
v11b	Welfare State	Issue
v11c	Education and Culture	Issue
v11d	Security and Army	Issue
v11e	Foreign Policy	Issue
v11f	Europe	Issue
v11g	Infrastructure	Issue
v11h	Society	Issue
v11i	Environmental Policies	Issue
v11j	Immigration	Issue
v11k	Labour Market and Employment	Issue
v12	Conflict	Style
v13	Language of Games, Sports and War	Style
v14	Reference to EU	Other
v15	Social Media	Other
v16	Fake News	Other
v17	Polling Institutes	Other
v18	Scandals	Other
v19a	Positive Sentiment - Rauh	Sentiment
v19b	Negative Sentiment - Rauh	Sentiment
v19c	Overall Sentiment - Rauh	Sentiment
v20a	Positive Sentiment - SentiWS	Sentiment
v20b	Negative Sentiment - SentiWS	Sentiment

Variable	Label	Type
v20c	Overall Sentiment - SentiWS	Sentiment
v21a	Personalization SPÖ	Personalization
v21b	Personalization ÖVP	Personalization
v21c	Personalization FPÖ	Personalization
v21d	Personalization Greens	Personalization
v21e	Personalization NEOS	Personalization
v21f	Personalization JETZT – Pilz List	Personalization

4.2. Metadata

Metadata were derived from the source texts automatically by AmCAT, and when necessary were manually corrected or recoded.

- ***Article ID* (v0):** ID number by which the article can be recognized and searched within AmCAT.
- ***Headline* (v1):** Headline of the news report, stored as a string variable.
- ***Medium* (v2_l1-v2_l4):** Reported on four different levels. Level 1 (v2_l1) reports the type of media (TV, print, online, radio), level 2 (v2_l2) aggregates media outlets to subclasses (e.g., Tabloids) and level 3 (v2_l3) reports the specific media outlet in which the article was published. For TV and radio reports, and additional variable (v2_l4) provides information on the specific news show a report appeared in. For an overview on all included media outlets and their corresponding values and labels on each level, see *Table 3* and *Table 4*.

Table 3. Medium (v2) - Level values and labels

L1 Value	L1 Label	L2 Value	L2 Label	L3 Value	L3 Label
1000	TV	1100	Public Service TV	1110	ZIB 1
				1120	ZIB 2
				1130	ZIB (other)
				1140	Regional News (TV)
		1200	Commercial TV	1210	ATV
				1220	Puls 4
2000	Print	2100	Quality Print	2110	Der Standard
				2120	Die Presse
				2130	Salzburger Nachrichten
		2200	Tabloid Print	2210	Kronen Zeitung
				2220	Österreich
				2230	Heute
		2300	Mid-market Print	2310	Kurier
				2320	Kleine Zeitung

L1 Value	L1 Label	L2 Value	L2 Label	L3 Value	L3 Label
3000	Online	3100	Quality online	3110	derstandard.at
				3120	diepresse.com
				3130	sn.at
		3200	Tabloid online	3210	krone.at
				3220	oe24.at
				3230	heute.at
		3300	Mid-market online	3310	orf.at
				3320	kurier.at
				3330	kleinezeitung.at
4000	Radio	4100	Public Service Radio	4110	Ö1
				4120	Regional News (Radio)
				4130	Ö3

Table 4. Media (v2) Level 4 for TV and radio.

L3 Value	L3 Label	L4 Value	L4 Label
1110	ZIB 1	1111	ZIB 1
1120	ZIB 2	1121	ZIB 2
		1122	ZIB 2 am Sonntag
		1123	ZIB 2 Spezial
1130	ZIB (other)	1131	ZIB
		1132	ZIB Spezial
		1133	ZIB Nacht
		1134	ZIB Nacht Spezial
		1135	ZIB Flash
1140	Regional News (TV)	1141	Wien Heute (Vienna)
		1142	Niederösterreich Heute (Lower Austria)
		1143	Oberösterreich Heute (Upper Austria)
		1144	Steiermark Heute (Styria)
		1145	Tirol Heute (Tyrol)
		1146	Kärnten Heute (Carinthia)
		1147	Salzburg Heute (Salzburg)
		1148	Vorarlberg Heute (Vorarlberg)
		1149	Burgenland Heute (Burgenland)
1210	ATV	1211	ATV Aktuell 17:20
		1212	ATV Aktuell 18:20
		1213	ATV Aktuell 19:20
1230	Puls 4	1231	Puls 4 News 08:00
		1232	Puls 4 News 18:45
		1233	Puls 4 News 20:00
4110	Ö1	4111	Ö1 Morning News 07:00
		4112	Ö1 Morning News 08:00
		4113	Ö1 Midday News 12:00
		4114	Ö1 Evening News 18:00
		4115	Ö1 News (Other)
4120	Regional News (Radio)	4121	Radio Wien News (Vienna)

L3 Value	L3 Label	L4 Value	L4 Label
		4122	Radio Niederösterreich News (Lower Austria)
		4123	Radio Oberösterreich News (Upper Austria)
		4124	Radio Steiermark News (Styria)
		4125	Radio Tirol News (Tyrol)
		4126	Radio Kärnten News (Carinthia)
		4127	Radio Salzburg News (Salzburg)
		4128	Radio Vorarlberg News (Vorarlberg)
		4129	Radio Burgenland News (Burgenland)
4130	Ö3	4131	Ö3 Morning News 07:00
		4132	Ö3 Midday News 12:00
		4133	Ö3 News (Other)

- ***Medium Included (v3):*** Whether the media outlet (*v2_l3* for print and online, *v2_l4* for TV and radio) was included in the data sets for the elections of 2013 (*v3_13*) or 2017 (*v3_17*).
- ***Date* (v4):** The date on which the article was published.
- ***Page* (v5):** The page on which the article was published. Only reported for articles in print media (when available).
- ***Section* (v6):** Automatically coded variable on the section in which the article was published. Differentiating section names were harmonized to the following section labels in *Table 5*. A section was only assigned when possible. Sections are only available for print and online media.

Table 5. Section (v6) - Values and labels

Value	Label	Value	Label
1	Title Page	11	Science and Health
2	Domestic Politics	12	Sports
3	Topic of the Day	13	Letter to the Editor
4	Local News	14	Opinion Piece
5	Foreign Affairs	15	TV/Radio-Programs
6	Feuilleton	16	Events and Activities
7	Economy	17	Reportage and Features
8	Culture	33	Other
9	Communication and Media	99	Not assignable
10	Society		

- ***Word count* (v7):** Length of main text of articles, coded automatically.

4.3. Parties

The appearance in news reports of the main parties in the 2019 national election (SPÖ, ÖVP, FPÖ, Greens, NEOS, JETZT - Pilz List) was coded with the following search strings. All party variables have the value 1 when the given party is mentioned in the news report, and 0 if it is not mentioned.

- ***SPÖ* (v8a):**

```
spo sp "partei sozialdemo*"~5 sozialdemokrat* "die roten" "den roten" sozi sozialist*
```

[Precision: 0.94 | Recall: 0.94]

- ***ÖVP* (v8b):**

```
ovp vp volkspartei "liste kurz"~5 "neue* volkspartei" "die schwarzen" "den schwarzen" "die turkisen" "den turkisen"
```

[Precision: 0.94 | Recall: 0.93]

- ***FPÖ* (v8c):**

```
fpo freiheitlich* fp "partei freiheitlich*"~5
```

[Precision: 0.94 | Recall: 0.92]

- ***Greens* (v8d):**

```
(("grune partei" "die grunen" "die grune" "der grunen" "der grune" "die grune partei" "grune alternative" okopartei grunpolitiker* "grun politiker*" "den grunen" "tiroler grune*" "wiener grune*" "grune* tirol" "grune* wien" "grune* burgenland" "burgenland* grune*" "grune* karnten" "karntner grune*" "grune* salzburg" "salzburger grune*" "grune* oberosterreich" "oberosterreichische* grune*" "grune* steiermark" "steirische* grune*" "grune* vorarlberg" "vorarlberger grune*" "grune* niederosterreich*" "niederosterreichische* grune*" "grune* spitzenkandidat*"~5 "grune* vorstandssprecher*"~5 "grune* landtagsabgeordnet*"~5 "grune* fpo*"~5 "grune* freiheitl*"~5 "grune* blau*"~5 "grune* spo"~5 "grune* sp"~5 "grune* sozialdemokrat*"~5 "grune* sozialist*"~5 "grune* rot*"~5 "grune* ovp"~5 "grune* vp"~5 "grune* volkspartei"~5 "grune* schwarz*"~5 "grune* neos"~5 "grune* kpo"~5 "grune* kommunist*"~5 "grune* chef*"~5 "grune* vorsitz*"~5 "grune* felipe"~5 "grune*
```

```
kogler"~5 "grune* sprecher*"~5 "grune* bundesvorstand*"~5
"grune* bundessprecher*"~5 "grune* landesvorstand*"~5
"grune* parlament*"~5 "grune* regierung*"~5 "grune*
opposition*"~5 "grune* partei*"~5 "grune* bundnis*"~5
"grune* fraktion*"~5 "grune* klub*"~5 "grune* vize*"~5
"grune* burgermeister"~5 "grune* stadtrat*"~5 "grune*
landesrat*"~5 "grune* bereichssprecher*"~5 "grune*
referent*"~5 "grune* landespart*"~5 "grune* ortsgrupp*"~5)
NOT("grune* (deutsch* cdu csu fdp jamaika spd afd ozdemir)"~7
"bundnis 90 grune*"~5 "grune* mark")
```

[Precision: 0.87 | Recall: 0.89]

- ***NEOS* (v8e):**

```
neos "das neue osterreich"
```

[Precision: 0.90 | Recall: 0.95]

- ***JETZT - Pilz List* (v8f):**

```
"liste jetzt" "jetzt (pilz stern holzinger vogtenhuber
balluch schroeder)"~2 "liste pilz"~5
```

[Precision: 0.84 | Recall: 0.84]

4.4. Main candidates

The appearance in news reports of the main candidates in the 2019 national election was coded with the following search strings. A detailed account on the validation of these search strings can be found in the following section on the validation procedure. All variables have the value 1 when the given candidate is mentioned in the news report, and 0 if he/she is not mentioned.

- ***Pamela Rendi-Wagner - SPÖ* (v9a):**

```
"rendi wagner*"
```

[Precision: 0.97 | Recall: 0.97]

- ***Sebastian Kurz - ÖVP*** **(v9b):**

```
"sebastian kurz"~4 "kurz (kanzler bundeskanzler ovp vp volkspartei parteiobmann spitzenkandidat)"~5
```

[Precision: 0.94 | Recall: 0.91]

- ***Norbert Hofer - FPÖ*** **(v9c):**

```
"norbert hofer*" ("hofer* (fpo freiheitlich* chef spitzenkandidat* partei* kickl strache rendi reisinger kurz pilz fpo spo ovp neos wahl wahlkampf duell)"~5 NOT("thomas hofer"))
```

[Precision: 0.96 | Recall: 0.93]

- ***Werner Kogler - Greens*** **(v9d):**

```
"werner kogler*" "kogler* (grune gruner grunen chef spitzenkandidat* partei* rendi reisinger hofer kurz pilz fpo spo ovp neos wahl wahlkampf duell)"~5
```

[Precision: 0.94 | Recall: 0.99]

- ***Beate Meinl-Reisinger - NEOS*** **(v9e):**

```
"meinl reisinger*"
```

[Precision: 0.91 | Recall: 0.96]

- ***Peter Pilz - JETZT - Pilz List*** **(v9f):**

```
"pilz peter"~15
```

[Precision: 0.95 | Recall: 0.89]

4.5. Government

Due to the successful vote of no confidence against the Government Kurz I in the Austrian parliament, Austria was governed by a cabinet of experts under Chancellor Brigitte Bierlein in the lead-up to the elections of 2019. To account for the relevance of this government of experts in the media coverage, *v10* measures whether the expert government or a specific minister of this cabinet was mentioned in a media report (1) or not (0).

- ***Government of Experts*** **(v10):**

```
expertenregierung* beamtenregierung* "(experten beamten bierlein) (regierung kabinett minister*)" bierlein* jabloner* "(eduard bundesminister minister finanzminister sportminister beamtenminister) muller" schallenberg zarfl rauskala "udolf strobl" peschorn starlinger patek reichhardt stilling
```

[Precision: 0.98 | Recall: 0.95]

4.6. Issues

We measure the appearance of issues by aggregating search string results on the finest level to the most general level that is reported in our data set. Issues were only validated on the most general level. Therefore, we recommend validating issue search strings on more fine-grained levels when needed. Due to the extensive number of search strings, only precision and recall values as well as the amount of search strings used for one issue are reported in this documentation. For all issues, the variable takes the value 1 if the issue is mentioned and 0 if it is not mentioned in the article.

- ***Economy, Budget and Taxes*** **(v11a):**

Aggregates the results of 113 search strings, covering the following topics: Economic policies, liberalization, deregulation, globalization, financial markets and banks, business policies, agriculture, tourism, state budget, government debt, and taxation policies.

[Precision: 0.81 | Recall: 0.85]

- ***Welfare State*** **(v11b):**

Aggregates the results of 101 search strings, covering the following topics: Welfare policies and spending, health system, care, pensions, family and children support, housing, and sports.

[Precision: 0.87 | Recall: 0.70]

- ***Education and Culture*** **(v11c):**

Aggregates the results of 66 search strings, covering the following topics: Educational policies, schools, universities, research, vocational education, adult education, and cultural policies.

[Precision: 0.78 | Recall: 0.84]

- ***Security and Army*** **(v11d):**

Aggregates the results of 71 search strings, covering the following topics: Crime, penal system, police, terrorism, and military policies.

[Precision: 0.87 | Recall: 0.82]

- ***Foreign Policy*** **(v11e):**

Aggregates the results of 17 search strings, covering the following topics: International cooperation, international and bilateral relations and issues on war and peace.

[Precision: 0.78 | Recall: 0.79]

- ***Europe*** **(v11f):**

Aggregates the results of 79 search strings, covering the following topics: EU integration, EU treaties, EU extensions and exits, EU policies, EU participation and the Euro.

[Precision: 0.91 | Recall: 0.87]

- ***Infrastructure*** **(v11g):**

Aggregates the results of 38 search strings, covering the following topics: Road, rail and air traffic, shipping, energy policies, telecommunication, digitalization, and infrastructure investments and policies.

[Precision: 0.90 | Recall: 0.76]

- ***Society*** **(v11h):**

Aggregates the results of 76 search strings, covering the following topics: Gender equality, marriage, sexual orientation, abortion, bioethics, religion, censorship, handling of Austria's National Socialist Past, far-right extremism, anti-Semitism, patriotism, nationalism, minority rights and media policies.

[Precision: 0.86 | Recall: 0.72]

- ***Environmental Policies*** **(v11i):**

Aggregates the results of 25 search strings, covering the following topics: Environmental protection, climate change and animal welfare.

[Precision: 0.96 | Recall: 0.85]

- ***Immigration*** **(v11j):**

Aggregates the results of 61 search strings, covering the following topics: Immigration, refugees, asylum, integration of immigrants, Islam, racism and anti-racism, citizenship policies, and the handling of immigrants concerning the welfare system, labour market, education, and crime.

[Precision: 0.88 | Recall: 0.77]

- ***Labour Market and Employment*** **(v11k):**

Aggregates the results of 44 search strings, covering the following topics: Labour market, unemployment, employee rights and welfare and consumer protection.

[Precision: 0.77 | Recall: 0.84]

4.7. Style

The following two search strings measure the appearance of certain stylistic elements in news reports. Both variables take the value 1 if the stylistic element is present, and 0 if not. A detailed account on the validation of these search strings can be found in the following section on the validation procedure.

- ***Conflict* (v12):**

Our operationalization of a conflict frame includes cases where at least one of the following indicator questions applies: (1) *Does the item reflect disagreement between parties, individuals, groups or countries?* OR (2) *Does one party, individual or group insult/offend/accuse another?* Note that we do not code cases that refer to two or more sides of the problem under our operationalization of a conflict frame. The following search string was used to measure conflict:

```
((kritik* NOT("kaum kritik")) kritisier* kampfansage (streit* NOT("außer streit" "streitmacht" "kein* streit")) gestritt* gegner* unredlich* unfahig* konflikt konflikte (angriff* NOT("in Angriff nehmen" "in Angriff genommen")) "nicht akzept*"~5 attack* heftige* (protest* NOT(protestwähler)) offensive angreifen angegriffen "griff* an"~10 "greift an"~10 vollholler unvereinbarkeit* Eklat tumult* strittig* intervention* intervenier* "wehr* gegen"~5 "vorwurf gegen*"~10 "warf vor"~10 "wirft vor"~10 frontalangriff* (widerspricht NOT("widerspricht nicht")) (widersprach NOT("widersprach nicht")) (widersprochen NOT("nicht widersprochen")) angegriffen* gezerr* schimpf* geschimpft*) AND NOT (ostukraine donbass syrien* syrisch* agypt* djihad* iran* sudan* sudsudan* nordkorea* un uno saudi* staatengemeinschaft* sicherheitsraat* terroris* kaida qaida pakistan sudan* jemen* libyen lybisch* somali* eritre* kongo* elfenbeinkust* ivor* ivoire hamas hisbollah miliz gotteskrieger taliban* is "islamische* staat" isis)
```

[Precision: 0.82 | Recall: 0.63]

- ***Language of Games, Sports and War* (v13):**

This variable captures whether a news report features terms of games, sports or war to describe actions and interaction between political actors or the election itself. The following search string was used to measure the appearance of language of games, sports and war:

```
((agressiv* durchpeitsch* zerstor* "greift an"~5 "griff an"~5 "finish" vs "Aufholjagd" "Nachspiel" show* kill* endspurt schlagabtausch verspiel* zweikampf* flugelkampfe kraftakt "presch* vor" dreikampf* angriff* angegriff* angreif* vergeltung* nahkampf* wortgefecht* scharmutzel* gefecht* krieg kriege kriegerisch* arsenal* giftschrank* aufmarsch*
```

gegenschlag* uberraschungsangriff* uberraschungsschlag* armada bombig* bombadier* bombardier* explosiv* propaganda* neutralisier* konter kontern gekontert kontert* konterattack* konterangriff* feldzug* sprengkraft sprengstoff sprengt* schlachtfeld* offensiv* defensiv* hinterhalt* schlacht hauptquartier (kampf* NOT("kampf* tranen"~5)) gekampft* kanone* (schiess* NOT(schiesserei*)) geschoss* zuschlagen zugeschlagen* zuschlug zuschlagt "anschlag* auf" kommando* geheimwaffe* belager* krise* regierungskrise* koalitionskrise* attacke* wahlschlacht* himmelfahrtskommando* kreuzzug* blitzkrieg* verteidigung verteidigt* schutzengraben* geschutz mobilisier* erober* vormarsch* bekampf* keule verbotskeule richtungskampf* putsch* widerstand* feind* mitstreiter* rebellisch* aufstand aufruhr* intervention* revolte* palastrevolte* gerangel krawall* scharmutzel* konfrontation* konfrontier* frontalangriff* kampagne* kampagnisier* aufbegehr* einschiess* eingeschoss* entscharf* stichel* heckenschutze* abschiess* abgeschoss* (abschuss NOT("abschuss tier*"~20)) querschiess* querschuss* querschutze* (front* NOT(frontex "front national" frontalkollision "front* auto*"~10 "front* fahrzeug*"~10 "front* unfall*"~10 "front* kolli*"~10)) koalitionsfront* regierungsfront* oppositionsfront* grabenkampf* gelandegewinne vorstoss* deckung* storfeuer* durchbruch* abruck* ruckzugsgefecht* verluste* ruckzug vordringen* anfuhrer "fuhrung ausbauen"~5 "in fuhrung"~5 positionspapier* bollwerk* manover* taktik* taktisch* wahltaktisch* strategie* strategisch* wahlstrategi* kapitulation* kapitulier* "flucht nach vorn" fluchtet* stormanover* giftpfeil* blockad* blockier* mobilisier* rekrutier* flankier* ruckendeck* "rucken gedeckt" mitstreiter* bollwerk* schlachtschiff* machtkampf* kampfkandidat* kampfabstimm* uberflugel* vorbeizieh* vorbeigezog* tauchstation* abtauch* abgetaucht* favorit favoritin aussenseiter* outsider* geheimfavorit* rochade* personalrochade* rochier* todesspirale* foul* finale* angezahlt* angeschlagen* triumph* pyhrrus* steilpass* steilvorlage* matchwinner* eigentor* powerplay arena wahlkampfarena duell* kanzlerduell* gladiator* auferstehung* opfergang* aufstachel* aufgestachel* fair* unfair* gewinn* sieger* verlierer* schlappe uberrunde* bruchlandung* angeschlagen* angezahlt* startschuss* startrampe* startlocher* "sturm auf" "ansturm auf" "war room" "aus volle* rohr*" "ins abseits" "im abseits" "rote karte" "auflage verwandel*" "vorlage verwandel*" "fest im sattel" "sicher im sattel" "auf der strecke bleib*" "blieb auf der strecke" "auf der strecke geblieb*" "kopf an kopf" "ins ziel" "ins spiel" "in stellung" "stellung ein*" "stellung beziehen" "bezog*

```
stellung" "stellung halten" "stellung gehalten" "hielt stellung"~3) NOT (syrien* assad syrisch* agypt* djihad* iran* sudan* sudsudan* nordkorea* un uno staatengemeinschaft* sicherheitsraat* terroris* kaida qaida pakistan* sudan* jemen* libyen lybisch* somali* eritre* kongo* elfenbeinkust* ivor* ivoire hamas hisbollah miliz gotteskrieger taliban* terrormiliz "IS-terrormiliz" "IS" "kampfjets" bombardier "ISIS" terroranschlag* "terror anschlag" "todlich* anschlag" terrorakt terrorgefahr terrorregime* terrormiliz terrorwarnstufe* terrororganisation*))
```

[Precision: 0.71 | Recall: 0.81]

4.8. Other

Next to the variables capturing actors, issues and stylistic features, the appearance of the following five items was coded. All variables take the value 1 if the news report contains the specific item, and 0 if not.

- ***Reference to EU*** **(v14):**

Measures whether a news report contains an explicit reference to EU law, policies, regulations and measures using the search string below. Note that this measurement differs from the definition used in the 2017 data set, where the variable also included references to the European Union itself or specific institutions.

```
("(eu europ*) (deal richtlinie* gesetz* vorschrift* vorgabe* regelung* genehmigung regel geregelt recht rechtlich* primarrecht* sekundarrecht* abkommen freihandelsabkommen grundverordnung verpflichtung verordnung)"~6 "(eu europ* ezb eurozone) (geldpolitik schuldenpolitik zinspolitik haushaltsvorgab* haushaltsvorschrift* haushaltsregel* budgetvorgab* budgetvorschrift* budgetregel*)"~6 "(eu europ*) (fluchtlingspolitik asylpolitik zuwanderungspolitik migrationspolitik)" "eu (vertrag* verfassung)" "vertrag (lissabon amsterdam nizza)"~3 "eugh* (urteil* verurteil* klage)"~6 "europ* gerichtshof* (urteil* verurteil* klage)"~6 europarecht mercosur ttip ceta)
```

[Precision: 0.86 | Recall: 0.93]

- ***Social Media*** **(v15):**

Measures whether a news report contains an explicit reference to the Social Media, Social Media platforms or Social Media sources using the following search string:

```
(twitter* facebook* instagram* whatsapp* tweet* snapchat* pinterest* youtube* "social (media* network*)" "sozial* (medien* netz netze netzen netzwerk*)" posting postet* gepostet getwittert) NOT("teilen share on facebook tweet" "whatsapp-service")
```

[Precision: 0.99 | Recall: 0.99]

- ***Fake News*** **(v16):**

Measures whether a news report contains an explicit reference to the role and extend of fake news or the usage of fake news by political actors using the following search string:

```
"fake news" OR falschmeldung OR "alternativ* fa:t*" OR postfakt* OR "post fa:t*" OR lügenpresse OR desinformation OR medienbashing OR „medien bashing"~3 OR „foto* manipul*"~5 OR „bild* manipul*"~5 OR „information* manipul*"~5 „gef:lscht information*"~5 OR „verf:lsch* information*"~5 OR „f:lschlich* information*"~5 OR „unkorrekt* information*"~5 lügenpresse* „lüge* presse"~5 „lüge* medien"~5 „lüge* medium"~5 „lüge* zeitung"~5 „lüge* orf"~5 „gelogen medien"~5 „gelogen medium"~5 „gelogen presse"~5 „gelogen zeitung"~5 „gelogen orf"~5 „unehrlich* medien"~5 „unehrlich* medium"~5 „unehrlich* presse"~5 „unehrlich* zeitung"~5 „unehrlich* orf"~5 „unwahr* medien"~5 „unwahr* medium"~5 „unwahr* presse"~5 „unwahr* zeitung"~5 „unwahr* orf"~5 „erfunden* medien"~5 „erfunden* medium"~5 „erfunden* presse"~5 „erfunden* zeitung"~5 „erfunden* orf"~5 „verlogen* medien"~5 „verlogen* medium"~5 „verlogen* presse"~5 „verlogen* zeitung"~5 „verlogen* orf"~5
```

[Precision: 0.93 | Recall: 0.80]

- ***Polling Institutes*** **(v17):**

Measures whether a news report contains an explicit reference to a polling institute using the following search string:

```
(umfrag* wahlumfrag* sonntagsfrag* politbarometer politikbarometer) AND (ifes "integral meinungsforschung*" sora gallup imas "imas international" ogm "market institute" "unique research" "peter hajek" akonsult spectra "spectra markforschung" "research affairs" gfk marktagent imad)
```

[Precision: 1.00 | Recall: 0.89]

- ***Scandals*** **(v18):**

Measures whether a news report contains an explicit reference to a specific political scandal (e.g., Ibiza affair or "Schredder-Affäre") using the following search string:

```
(ibiza ibizaaffare ibizagate) (spesenaffare spesenskandal "(fpo strache*) (spesen* parteispesen* mietzuschuss*)"~10 "spesen* (affare* skandal* blau*)") ("(causa fall) chorherr"~2 "chorherr (spenden* spender*) (soravia ithuba)"~10 "chorherr (spenden spendenaffäre spendenskandal)"~5) (schreddern "schredder (affare gate skandal)"~2 "schredder (kurz ovp volkspartei vp festplatt* kanzler)"~20 schredderaffare schredderskandal) ("sidlo (vorstand* bestellung casinos novomatic)"~30 "casino* (causa skandal affare)") ("einzelfall* (fpo freiheitlich hofer strache rechtsextrem* nazi* nationalsozial*)"~10 rattengedicht* liederbuchaffare liederbuchskandal "liederbuch (affare skandal)" "causa stenzel" "stenzel identitäre*"~5 "(klinger* landesrat*) mischkultur*"~5) (spendenaffare spendenskandal hortengate wahlkampfkostenaffare "(wahlkampfkosten wahlkampfausgaben wahlkampfspenden spenden konto konten buchhaltung) (illegal* verdeckt* verschleier* doppelt* enthull*) (ovp vp volkspartei falter)"~15) ("(ovp vp volkspartei) (hacker* cyber*)"~5 "(hacker datenklau datenleck) (angriff affare skandal angriff)" hackeraffare) ("bvt (skandal* affare*)"~5)
```

[Precision: 0.98 | Recall: 0.90]

4.9. Sentiment

To calculate sentiment-scores on article level, we used two existing dictionaries for German language: A sentiment dictionary for German political language by Christian Rauh (Rauh,

2018[2]) and the more general sentiment dictionary SentimentWortschatz/SentiWS (Remus et al., 2010; v.2.0[3]). Both dictionaries attribute positive and negative sentiments to certain cue words. The overall score of negativity or positivity was normalized by the number of words of the given article. Note that numbers and punctation were removed from the texts as it is assumed that these do not bear any sentiment.

For both dictionaries, we provide a separate normalized score of positivity and negativity together with an overall normalized sentiment score. The normalized positivity score theoretically ranges from 0 (no positive sentiment cues) to 1 (all words of the text contain positive sentiment), whereas the normalized sentiment score ranges from 0 (no negative sentiment cues) to -1 (all words of the text contain negative sentiment). As the overall normalized sentiment score is calculated as the sum of the two separate sentiment scores, it theoretically ranges from -1 to 1. As both dictionaries have already been validated by their creators, we did not conduct additional validation of sentiment scores. The following variables in the data set capture the sentiment of the news reports:

- ***Sentiment Rauh* (v19a-c):** Normalized sentiment scores calculated with the sentiment dictionary for German political language (Rauh, 2018). We provide information on sentiments in three ways: Positive sentiment – Rauh (*v19a*), Negative sentiment – Rauh (*v19b*) and Overall sentiment – Rauh (*v19c*).
- ***Sentiment SentiWS* (v20a-c):** Normalized sentiment scores calculated with the SentiWS dictionary (Remus et al., 2010; v.2.0). We provide information on sentiments in three ways: Positive sentiment – SentiWS (*v20a*), Negative sentiment – SentiWS (*v20b*) and Overall sentiment – Rauh (*v20c*).

4.10. Personalization

Personalization is operationalized as a continuous variable that represents the ratio of the summed mentions of a party's main candidate to the sum of main candidate and party

[2] Rauh, C. (2018). *Replication Data for: Validating a sentiment dictionary for German political language* (DRAFT VERSION). Harvard Dataverse. https://doi.org/10.7910/DVN/BKBXWD

[3] Remus, R., Quasthoff, U., & Heyer, G. (2010). *SentiWS-A Publicly Available German-language Resource for Sentiment Analysis.* LREC. http://www.lrec-conf.org/proceedings/lrec2010/pdf/490_Paper.pdf

mentions per article. It therefore ranges between 0 (only party is mentioned) to 1 (only main candidate is mentioned). Note that personalization is not coded (NA) if neither party nor main candidate is mentioned in a media report. As party and main candidate search strings were already validated separately (see Section 4.3 and 4.4), we do not provide validation for personalization scores. We provide personalization scores for the following six parties and their respective main candidates:

- ***Personalization SPÖ*** **(v21a)**
- ***Personalization ÖVP*** **(v21b)**
- ***Personalization FPÖ*** **(v21c)**
- ***Personalization Greens*** **(v21d)**
- ***Personalization NEOS*** **(v21e)**
- ***Personalization JETZT - Pilz List*** **(v21f)**

5. Validation Procedure

5.1. Overview

Following face validity checks during the search string creation and adjustment process (see Section 4.1) we performed a final validation of our automatic methods by comparing their results to manual coding. This comparison results in a following binary contingency table:

	Manual = 1	**Manual = 0**	
Automatic = 1	True Positive (A)	False Positive (B)	*Predicted Positives* (A+B)
Automatic = 0	False Negative (C)	True Negative (D)	*Predicted Negatives* (C+D)
	Real Positives (A+C)	*Real Negatives* (B+D)	

Based on this comparison, we calculated to measures for validation: Precision and recall. Precision refers to the ratio of true positive results to all predicted positive results of a search string [P=A/(A+B)]. Recall refers to the ratio of true positive results of a search string to all real positives [R=A/(A+C] (Powers, 2011[4]).

[4] Powers, D. M. (2011). Evaluation: From Precision, Recall and F-Measure to ROC, Informedness, Markedness & Correlation. *Journal of Machine Learning Technologies*, 2(1): 37–63. https://arxiv.org/abs/2010.16061

For the variables on parties and main candidates we were able to draw on the manually coded data from the AUTNES Manual Content Analysis of the Media Coverage 2019 (N=6689). For all other variables, we conducted additional validation coding. We followed the proposed validation scheme of Stryker et al. (2006[5]). Based on the precision and recall values of our face validity checks we calculated the minimum number of coded articles needed to sufficiently validate our search strings. This resulted in a validation data set of 1000 articles (AmCAT set ID: 5543) to validate the missing items. The four student coders that coded our additional validation set were extensively trained, receiving in total three days of coder training. To test intercoder reliability, at the end of the training we conducted reliability tests that can be found in the following subsection (Section 5.2).

Furthermore, note that we refrained from validating all the sentiment variables, as the two sentiment dictionaries have already been validated by their creators. Personalization variable did not require any validation, as the party and candidate variables used to calculate personalization scores, are validated on their own. Finally, note that we do not provide validation for metadata variables (v0-v8), as these variables were captured automatically from the information provided by the respective data sources.

Table 6 shows how each variable was validated. If it was validated via data from AUTNES Manual Content Analysis of the Media Coverage 2019 (doi: 10.11587/FVTDXG), the corresponding variable of the manual data set is given.

Table 6. Validation procedure

Variable	Label	Validation data	AUTNES manual variable
v8a	SPÖ	AUTNES manual	v14a (recoded)
v8b	ÖVP	AUTNES manual	v14c (recoded)
v8c	FPÖ	AUTNES manual	v14e (recoded)
v8d	Greens	AUTNES manual	v14g (recoded)
v8e	NEOS	AUTNES manual	v14i (recoded)
v8f	JETZT - Pilz List	AUTNES manual	v14k (recoded)
v9a	Pamela Rendi-Wagner	AUTNES manual	v14b (recoded)
v9b	Sebastian Kurz	AUTNES manual	v14d (recoded)
v9c	Norbert Hofer	AUTNES manual	v14f (recoded)

[5] Stryker, J. E., Wray, R. J., Hornik, R. C., & Yanovitzky, I. (2006). Validation of Database Search Terms for Content Analysis: The Case of Cancer News Coverage. Journalism & Mass Communication Quarterly, 83(2), 413–430. https://doi.org/10.1177/107769900608300212

Variable	Label	Validation data	AUTNES manual variable
v9d	Werner Kogler	AUTNES manual	v14h (recoded)
v9e	Beate Meinl-Reisinger	AUTNES manual	v14j (recoded)
v9f	Peter Pilz	AUTNES manual	v14l (recoded)
v10	Government of Experts	Additional Validation	
v11a	Economy, Budget and Taxes	Additional Validation	
v11b	Welfare State	Additional Validation	
v11c	Education and Culture	Additional Validation	
v11d	Security and Army	Additional Validation	
v11e	Foreign Policy	Additional Validation	
v11f	Europe	Additional Validation	
v11g	Infrastructure	Additional Validation	
v11h	Society	Additional Validation	
v11i	Environmental Policies	Additional Validation	
v11j	Immigration	Additional Validation	
v11k	Labour Market and Employment	Additional Validation	
v12	Conflict	Additional Validation	
v13	Language of Games, Sports and War	Additional Validation	
v14	Reference to EU	Additional Validation	
v15	Social Media	Additional Validation	
v16	Fake News	Additional Validation	
v17	Polling Institutes	Additional Validation	
v18	Scandals	Additional Validation	

5.2. Intercoder Reliability

Table 7 shows the intercoder reliability of the additional validation and presents both Krippendorff's Alpha and Percentage Agreement for each variable. The data set for the inter-coder reliability tests consisted of 165 articles (AmCAT set IDs: 5319, 5359), oversampled for rarely occurring variables such as Fake News (*v16*). For intercoder reliability on parties and candidates, consult the documentation of the AUTNES Manual Content Analysis of Media Coverage 2019 (doi: 10.11587/FVTDXG).

Table 7. Intercoder reliability - Additional validation

Variable	Label	Krippendorff's Alpha	Percentage Agreement
v10	Government of Experts	0.959	0.98
v11a	Economy, Budget and Taxes	0.667	0.80
v11b	Welfare State	0.805	0.89
v11c	Education and Culture	0.583	0.91
v11d	Security and Army	0.680	0.80
v11e	Foreign Policy	0.525	0.97
v11f	Europe	0.724	0.88

Variable	Label	Krippendorff's Alpha	Percentage Agreement
v10	Government of Experts	0.959	0.98
v11g	Infrastructure	0.777	0.89
v11h	Society	0.530	0.73
v11i	Environmental Policies	0.901	0.94
v11j	Immigration	0.805	0.93
v11k	Labour Market and Employment	0.670	0.91
v12	Conflict	0.653	0.67
v13	Language of Games, Sports and War	0.441	0.50
v14	Reference to EU	0.738	0.96
v15	Social Media	0.769	0.94
v16	Fake News	0.712	0.89
v17	Polling Institutes	0.936	0.98
v18	Scandals	0.790	0.84

5.3. Results

Table 8. Validation - Precision and Recall

Variable	Label	Precision	Recall
v8a	SPÖ	0.94	0.94
v8b	ÖVP	0.94	0.93
v8c	FPÖ	0.94	0.92
v8d	Greens	0.87	0.89
v8e	NEOS	0.90	0.95
v8f	JETZT - Pilz List	0.84	0.84
v9a	Pamela Rendi-Wagner	0.97	0.97
v9b	Sebastian Kurz	0.94	0.91
v9c	Norbert Hofer	0.96	0.93
v9d	Werner Kogler	0.94	0.99
v9e	Beate Meinl-Reisinger	0.91	0.96
v9f	Peter Pilz	0.95	0.89
v10	Government of Experts	0.98	0.95
v11a	Economy, Budget and Taxes	0.81	0.85
v11b	Welfare State	0.87	0.70
v11c	Education and Culture	0.78	0.84
v11d	Security and Army	0.87	0.82
v11e	Foreign Policy	0.78	0.79
v11f	Europe	0.91	0.87
v11g	Infrastructure	0.90	0.76
v11h	Society	0.86	0.72
v11i	Environmental Policies	0.96	0.85
v11j	Immigration	0.88	0.77
v11k	Labour Market and Employment	0.77	0.84
v12	Conflict	0.82	0.63

Variable	Label	Precision	Recall
v13	Language of Games, Sports and War	0.71	0.81
v14	Reference to EU	0.86	0.93
v15	Social Media	0.99	0.99
v16	Fake News	0.93	0.80
v17	Polling Institutes	1.00	0.89
v18	Scandals	0.98	0.90

6. Data Set

In this section, we present a short overview over all variables in our dataset. To maintain clarity, recoded variables are shown in their most aggregated form. In addition to frequency tables, a short nominal description is given for each variable.

6.1. v6 Section

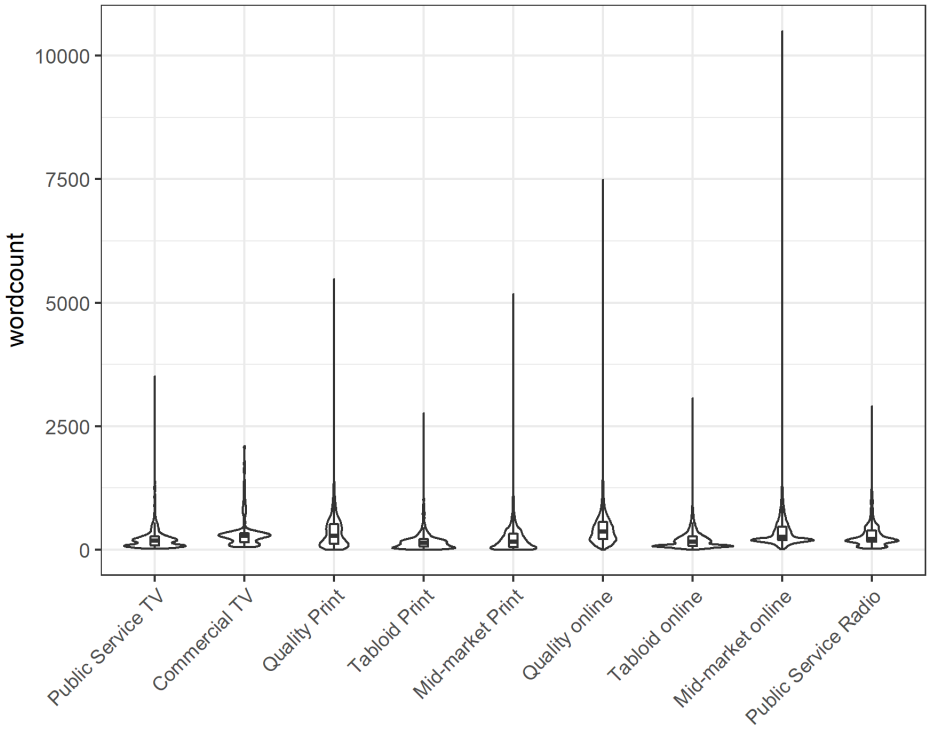
This variable was coded automatically and harmonized. It identifies the section in which an article was published.

	Public Service TV	Commercial TV	Quality Print	Tabloid Print	Mid-market Print	Quality Online	Tabloid Online	Mid-market online	Radio	Overall
Title Page			6.7	4.3	3.8					0.9
Domestic Politics			28.5	19.0	22.4	12.1		37.0		17.8
Topic of the day			11.5	4.2	2.3					1.0
Local news			4.8	36.8	26.1	0.9	5.3	0.3		6.1
Foreign Affairs			3.8	0.3	1.2	3.6		2.0		1.6
Feuilleton			5.7	0.4	15.9	0.5		9.2		4.2
Economy			6.4	0.7	2.4	0.8	0.2	2.2		1.4
Culture			1.3	0.5	2.7	0.5	0.1	1.4		0.8
Communication & Media			0.6		0.7	0.9	0.2			0.3
Society			0.1	0.7	2.4	0.1	1.1	0.6		0.6
Science & Health			0.2	0.2		0.2	0.2	0.4		0.2
Sports			1.4	1.4	0.8	1.6	13.7	14.3		7.6
Letter to the editor			0.8	0.0	3.3		0.2			0.3
Opinion piece			9.0	0.4	0.7	0.0		1.4		0.9
TV/Radio-Programs			1.1	0.5	0.2					0.1
Events and Activities				0.0	0.3			0.0		0.0
Reportage and feature			0.3	0.7	0.2					0.1
Other			4.7	0.1	0.9	1.5	0.7	1.3		1.1
Missing values	100	100	13.2	29.8	13.9	77.4	78.3	30.0	100	55.1
N	2384	186	2115	3544	2679	8586	8699	13511	2606	44310

Note: Percentage totals may be subject to rounding errors associated with individual components

6.2. v7 Word count

This variable was coded automatically. It counts the number of words each article contains.



	Mean	SD	Median
Public Service TV	254.27	332.45	183
Commercial TV	341.72	325.10	279
Quality Print	357.78	334.22	276
Tabloid Print	175.67	181.92	136
Mid-market Print	229.08	274.54	162
Quality online	445.05	411.71	365
Tabloid online	203.57	187.60	157
Mid-market online	364.58	308.89	257
Radio	303.00	294.79	212
Overall	315.28	318.79	222

6.3. v8a-f Parties

These variables measure whether a party is or is not mentioned in an article.

	Public Service TV	Commercial TV	Quality Print	Tabloid Print	Mid-market Print	Quality Online	Tabloid Online	Mid-market online	Radio	Overall
v9a – SPÖ	34.1	48.9	39.5	37.9	35.2	25.3	37.0	45.2	40.5	37.4
v9b – ÖVP	51.6	59.1	49.0	44.2	40.8	30.5	48.7	49.5	55.6	45.2
v9c – FPÖ	48.5	54.8	42.7	39.1	34.4	28.0	43.9	46.8	50.7	41.4
v9d – Greens	16.5	26.9	23.3	21.8	18.3	13.4	13.8	35.1	17.0	22.0
v9e – NEOS	17.5	22.6	21.2	14.4	12.2	11.9	7.4	33.4	21.8	19.2
v9f – JETZT – Pilz List	14.2	19.4	9.4	4.7	3.9	5.9	2.9	28.3	16.0	13.2
N	2384	186	2115	3544	2679	8586	8699	13511	2606	44310

Note: Percentage totals may be subject to rounding errors associated with individual components

6.4. v9a-f Main candidates & v10 Government of Experts

Measures whether a main candidate/the expert cabinet is or is not mentioned in an article.

	Public Service TV	Commercial TV	Quality Print	Tabloid Print	Mid-market Print	Quality Online	Tabloid Online	Mid-market online	Radio	Overall
v9a – Rendi-Wagner	9.0	11.3	11.3	13.0	7.6	5.5	4.3	4.2	6.8	6.2
v9b – Kurz	17.6	30.1	20.9	18.3	13.7	13.7	7.6	8.8	14.7	12.1
v9c – Hofer	11.5	12.4	13.4	12.6	8.9	6.8	4.9	5.3	9.0	7.3
v9d – Kogler	5.2	8.6	6.2	6.2	5.5	2.5	2.0	2.3	3.2	3.2
v9e – Meinl-Reisinger	5.4	7.5	5.9	5.5	4.5	2.5	1.7	2.0	3.5	3.0
v9f – Pilz	8.4	10.2	6.7	5.2	3.7	3.2	2.3	24.8	7.1	10.5
v10 – Government of Experts	19.5	16.7	9.9	6.5	7.6	9.4	3.1	7.8	19.3	8.5
N	2384	186	2115	3544	2679	8586	8699	13511	2606	44310

Note: Percentage totals may be subject to rounding errors associated with individual components

6.5. v11a-k Issues

These variables measure whether a specific issue is mentioned in a media report. Note that an article can include more than one issue.

	Public Service TV	Commercial TV	Quality Print	Tabloid Print	Mid-market Print	Quality Online	Tabloid Online	Mid-market online	Radio	Overall
v11a – Economy, Budget and Taxes	19.1	22.6	31.8	16.9	20.4	22.3	7.5	17.0	25.1	17.7
v11b – Welfare State	14.8	18.8	19.4	10.9	11.9	12.0	5.0	9.1	19.2	10.6
v11c – Education and Culture	9.9	10.8	22.1	9.6	17.5	18.9	5.7	12.8	14.5	13.0
v11d – Security and Army	17.7	15.1	18.8	12.0	11.7	18.2	8.4	13.8	18.9	14.1
v11e – Foreign Policy	3.9	1.6	6.9	2.9	2.6	6.1	1.5	4.3	4.3	4.0
v11f – Europe	15.6	17.2	15.7	7.8	7.7	14.9	4.3	10.7	16.6	10.7
v11g – Infrastructure	5.9	8.1	15.8	10.8	13.4	8.9	4.0	8.1	10.3	8.4
v11h – Society	15.5	11.3	17.5	8.2	10.2	16.0	5.1	8.7	15.4	10.7
v11i – Environmental Policies	14.8	22.0	20.1	14.3	15.0	12.9	6.0	10.8	16.1	11.8
v11j – Immigration	6.2	7.0	14.6	6.3	7.1	11.1	4.2	7.1	7.1	7.6
v11k – Labour Market and Employment	8.6	10.8	17.5	9.5	9.0	10.8	4.7	8.1	12.9	8.9
N	2384	186	2115	3544	2679	8586	8699	13511	2606	44310

Note: Percentage totals may be subject to rounding errors associated with individual components

6.6. v12-v18 Style and other variables

These variables measure whether an article features a certain element of style (conflict, language of games, sports and war) or if other specific topics (Reference to EU, social media, fake news, polling institutes, scandals) are mentioned in the article.

	Public Service TV	Commercial TV	Quality Print	Tabloid Print	Mid-market Print	Quality Online	Tabloid Online	Mid-market online	Radio	Overall
v12 – Conflict	27.4	34.4	31.1	22.8	22.6	36.9	19.0	27.8	28.7	27.4
v13 – Language of Games, Sports and War	32.4	39.8	42.8	36.8	33.6	53.8	46.7	49.3	30.3	45.3
v14 – Reference to EU	1.8	0.5	3.3	2.3	1.6	3.0	0.8	2.3	2.8	2.1
v15 – Social Media	6.5	14.5	9.4	5.8	5.3	13.0	10.4	10.2	4.8	9.6
v16 – Fake News	0.8	0.0	1.1	0.6	0.7	0.6	0.2	0.5	0.6	0.5
v17 – Polling Institutes	0.2	1.6	0.7	1.5	0.8	0.4	0.8	0.5	0.3	0.6
v18 – Scandals	27.9	30.1	20.7	15.8	13.5	13.9	7.0	10.0	21.8	13.1
N	2384	186	2115	3544	2679	8586	8699	13511	2606	44310

Note: Percentage totals may be subject to rounding errors associated with individual components

6.7. v19a-c Sentiment (Rauh)

These variables measure positive, negative and overall sentiment on article level based on the sentiment dictionary for German political language by Christian Rauh (Rauh, 2018[6]).

		Mean	SD	Min	Max
v19a – Positive Sentiment	Public Service TV	0.05396	0.02341	0	0.17857
	Commercial TV	0.05223	0.02263	0	0.12500
	Quality Print	0.05345	0.02299	0	0.21569
	Tabloid Print	0.05364	0.02295	0	0.18182
	Mid-market Print	0.05370	0.02437	0	0.18947
	Quality online	0.05393	0.02431	0	0.35294
	Tabloid online	0.05329	0.02344	0	0.23077
	Mid-market online	0.05373	0.02355	0	0.21739
	Radio	0.05222	0.02370	0	0.40000
	Overall	0.05358	0.02366	0	0.40000
v19b – Negative Sentiment	Public Service TV	-0.03742	0.02080	-0.14286	0
	Commercial TV	-0.03898	0.02086	-0.09551	0
	Quality Print	-0.03791	0.02072	-0.15152	0
	Tabloid Print	-0.03714	0.02142	-0.26667	0
	Mid-market Print	-0.03830	0.02263	-0.23077	0
	Quality online	-0.03745	0.02154	-0.25000	0
	Tabloid online	-0.03762	0.02216	-0.40000	0
	Mid-market online	-0.03733	0.02160	-0.25000	0
	Radio	-0.03754	0.02114	-0.20000	0
	Overall	-0.03751	0.02164	-0.40000	0
v19c – Overall Sentiment	Public Service TV	0.01654	0.03428	-0.13636	0.17857
	Commercial TV	0.01325	0.03364	-0.06180	0.12500
	Quality Print	0.01555	0.03382	-0.15152	0.16667
	Tabloid Print	0.01650	0.03448	-0.26667	0.18182
	Mid-market Print	0.01540	0.03674	-0.23077	0.18947
	Quality online	0.01648	0.03586	-0.25000	0.35294
	Tabloid online	0.01567	0.03544	-0.40000	0.23077
	Mid-market online	0.01640	0.03500	-0.25000	0.21739
	Radio	0.01468	0.03384	-0.15000	0.20000
	Overall	0.01607	0.03516	-0.40000	0.35294

[6] Rauh, C. (2018). *Replication Data for: Validating a sentiment dictionary for German political language* (DRAFT VERSION). Harvard Dataverse. https://doi.org/10.7910/DVN/BKBXWD

6.8. v20a-c Sentiment (SentiWS)

These variables measure positive, negative, and overall sentiment on article level based on the sentiment dictionary SentimentWortschatz/SentiWS (Remus et al., 2010; v.2.0[7]).

		Mean	SD	Min	Max
v20a – Positive Sentiment	Public Service TV	0.00436	0.00370	0.00005	0.04129
	Commercial TV	0.00431	0.00442	0.00010	0.04267
	Quality Print	0.00443	0.00370	0.00005	0.03577
	Tabloid Print	0.00435	0.00372	0.00004	0.05573
	Mid-market Print	0.00445	0.00424	0.00005	0.08372
	Quality online	0.00437	0.00386	0.00004	0.06993
	Tabloid online	0.00435	0.00372	0.00003	0.06004
	Mid-market online	0.00435	0.00382	0.00001	0.06958
	Radio	0.00418	0.00382	0.00007	0.07594
	Overall	0.00435	0.00382	0.00001	0.08372
v20b – Negative Sentiment	Public Service TV	-0.01176	0.01159	-0.35310	-0.00128
	Commercial TV	-0.01273	0.01389	-0.13241	-0.00211
	Quality Print	-0.01152	0.01210	-0.35310	-0.00104
	Tabloid Print	-0.01148	0.01084	-0.35310	-0.00095
	Mid-market Print	-0.01206	0.01029	-0.17655	-0.00056
	Quality online	-0.01213	0.01133	-0.35310	-0.00098
	Tabloid online	-0.01191	0.01040	-0.35310	-0.00095
	Mid-market online	-0.01193	0.01120	-0.35310	-0.00088
	Radio	-0.01190	0.01227	-0.35310	-0.00086
	Overall	-0.01191	0.01114	-0.35310	-0.00056
v20c – Overall Sentiment	Public Service TV	-0.00739	0.01225	-0.34910	0.02384
	Commercial TV	-0.00842	0.01500	-0.13091	0.01476
	Quality Print	-0.00709	0.01299	-0.34910	0.02097
	Tabloid Print	-0.00713	0.01180	-0.34910	0.03078
	Mid-market Print	-0.00761	0.01119	-0.17455	0.03681
	Quality online	-0.00776	0.01226	-0.34910	0.04558
	Tabloid online	-0.00756	0.01143	-0.34910	0.03262
	Mid-market online	-0.00758	0.01215	-0.34910	0.03195
	Radio	-0.00771	0.01302	-0.34910	0.02146
	Overall	-0.00755	0.01206	-0.34910	0.04558

[7] Remus, R., Quasthoff, U., & Heyer, G. (2010). *SentiWS-A Publicly Available German-language Resource for Sentiment Analysis.* LREC. http://www.lrec-conf.org/proceedings/lrec2010/pdf/490_Paper.pdf

6.9. v21a-f Personalization

These variables measure how personalized media coverage of a certain party is. It is computed by taking the ratio of main candidate mentions to the sum of mentions of each main candidate and his/her party.

v21a Personalization SPÖ

	Mean	SD	Min	Max
Public Service TV	0.094	0.174	0	0.8
Commercial TV	0.081	0.173	0	1
Quality Print	0.128	0.245	0	1
Tabloid Print	0.168	0.290	0	1
Mid-market Print	0.120	0.272	0	1
Quality online	0.087	0.199	0	1
Tabloid online	0.048	0.155	0	1
Mid-market online	0.036	0.129	0	1
Radio	0.059	0.151	0	1
Overall	0.070	0.188	0	1

v21b Personalization ÖVP

	Mean	SD	Min	Max
Public Service TV	0.133	0.233	0	1
Commercial TV	0.197	0.263	0	1
Quality Print	0.183	0.282	0	1
Tabloid Print	0.189	0.294	0	1
Mid-market Print	0.157	0.282	0	1
Quality online	0.181	0.283	0	1
Tabloid online	0.063	0.181	0	1
Mid-market online	0.063	0.175	0	1
Radio	0.100	0.207	0	1
Overall	0.108	0.230	0	1

v21c Personalization FPÖ

	Mean	SD	Min	Max
Public Service TV	0.081	0.174	0	1
Commercial TV	0.059	0.136	0	0.5
Quality Print	0.117	0.219	0	1
Tabloid Print	0.144	0.258	0	1
Mid-market Print	0.114	0.243	0	1
Quality online	0.082	0.185	0	1
Tabloid online	0.039	0.133	0	1
Mid-market online	0.037	0.127	0	1
Radio	0.060	0.146	0	1
Overall	0.064	0.170	0	1

v21d Personalization Greens

	Mean	SD	Min	Max
Public Service TV	0.092	0.182	0	1
Commercial TV	0.099	0.184	0	1
Quality Print	0.090	0.199	0	1
Tabloid Print	0.125	0.249	0	1
Mid-market Print	0.129	0.268	0	1
Quality online	0.053	0.145	0	1
Tabloid online	0.050	0.149	0	1
Mid-market online	0.019	0.094	0	1
Radio	0.047	0.111	0	1
Overall	0.050	0.154	0	1

v21e Personalization NEOS

	Mean	SD	Min	Max
Public Service TV	0.097	0.168	0	1
Commercial TV	0.089	0.134	0	0.364
Quality Print	0.136	0.251	0	1
Tabloid Print	0.210	0.327	0	1
Mid-market Print	0.215	0.343	0	1
Quality online	0.097	0.218	0	1
Tabloid online	0.105	0.227	0	1
Mid-market online	0.026	0.117	0	1
Radio	0.053	0.131	0	0.667
Overall	0.072	0.196	0	1

v21f Personalization JETZT - Pilz List

	Mean	SD	Min	Max
Public Service TV	0.322	0.327	0	1
Commercial TV	0.279	0.284	0	0.75
Quality Print	0.399	0.407	0	1
Tabloid Print	0.551	0.412	0	1
Mid-market Print	0.492	0.418	0	1
Quality online	0.309	0.362	0	1
Tabloid online	0.493	0.399	0	1
Mid-market online	0.449	0.209	0	1
Radio	0.216	0.287	0	1
Overall	0.417	0.288	0	1